

Smart Tag: Two Digital-Fabrication Prototypes for Produce Freshness Monitoring

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Abstract

Edible produce is often discarded because customers cannot easily judge whether a fruit is still suitable for consumption. We present *Smart Tag*, a digital-fabrication project framed around UN Sustainable Development Goal 12 (Responsible Consumption and Production) that addresses the problem through two complementary prototypes. The first is a connected retail prototype. An ESP32 reads an Adafruit BME688 environmental sensor and streams the readings wirelessly to a Raspberry Pi Zero W, which runs a Flask server, computes a freshness-aware discount, and updates a customer-facing e-ink shelf tag. We treat this version as an aspirational concept, since deploying a sensing tag in a real supermarket involves procurement, certification, and liability questions that go well beyond what a one-semester student project can answer. The second is a self-contained educational prototype. An ESP32-C3 drives a 16-pixel NeoPixel ring inside a translucent 3D-printed enclosure that signals fruit condition through colour, with no networking. This is the version we believe can be used today, in school canteens and classrooms, where it works both as a food-waste awareness object and as a teaching artefact for embedded electronics and digital fabrication. Both prototypes share a baseline-relative firmware philosophy validated through a controlled three-condition orange experiment that yields a roughly tenfold separation between fresh and overripe fruit in baseline-relative gas drop. The full project, including firmware, CAD, raw experimental data, analysis scripts, and step-by-step build instructions for both versions, is released as an open-source repository so that schools and other student projects can build on it directly.

Keywords

Prototype, Digital fabrication, 3D printing, Laser cutting, IoT, BME688, e-ink, NeoPixel, SDG 12, Food waste, Smart tag, Dynamic pricing

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1 Introduction

The brief for this project required us to ground our work in one of the United Nations Sustainable Development Goals. We chose SDG 12, Responsible Consumption and Production, because food waste is the SDG 12 sub-problem we could most directly address

with the digital-fabrication and embedded-sensing tools available to a student team. Edible produce is routinely discarded both at home and in retail because customers cannot easily judge whether a fruit is still pleasant to eat. The only freshness signal they usually see is a printed best-before date, which is conservative by design. Live measurements of humidity and spoilage-related gases can in principle support a different kind of signal: a real-time indication of the current condition of a particular fruit. This signal can feed freshness-aware discount logic in a retail setting, or it can simply tell someone looking at a fruit bowl whether the fruit is still worth eating.

We started with the obvious framing: a connected supermarket shelf tag with dynamic discounts, following prior work on dynamic pricing for perishables [2, 3]. As we built that version we became aware that the obstacles to retail adoption have very little to do with whether the sensor works. A supermarket would need to certify the device, integrate it into procurement and staff training, accept the liability of a sensor-driven freshness claim, and trust that the underlying signal is robust over many fruit varieties, ambient conditions, and supply-chain stages. None of these are problems we can solve in a one-semester student project, and we did not want to overstate what our prototype actually demonstrates.

We therefore built a second prototype, with a different intent and a different audience. The educational version is a stand-alone, network-free device that signals fruit condition through colour, and it serves two purposes at the same time. As a freshness signal, it makes food waste visible: a school canteen can see at a glance whether the fruit on display is still fresh, ripening, or past its best, without anyone having to read numbers off a display. As a teaching artefact, it is small, cheap, and well-documented enough that students can build it themselves as an introduction to embedded electronics, sensor calibration, and digital fabrication. In this framing, the retail prototype is the aspirational concept that motivated our exploration, and the educational prototype is the version we believe can be used in practice today.

To make the educational version usable beyond our own course, we released the full project as an open-source repository, including firmware for both prototypes, CAD and laser-cutter files, raw experimental data, the iteration history, and step-by-step build instructions. A school or maker space can clone the repository and either build the device as it is or use it as the starting point for their own iteration.

2 Related Work

Our project sits at the intersection of three threads of related work: dynamic pricing as a sustainability tool in the retail food sector,

IoT-based sensing of food spoilage, and real-time shelf-life criteria that replace fixed expiry dates with sensor-driven assessments. The four references discussed below shaped the framing of the project, the choice of sensor, and the way we present the device's output.

Sathyabama et al. [3] present an AI-driven expiry-prediction and dynamic-pricing system that supports the donation of products close to expiry, and argue that the approach can reduce waste while creating retailer value. The article framed dynamic pricing for us as both a sustainability lever and a commercial mechanism, and motivated the discount logic in our retail prototype. Building on that idea, Chung [2] discusses the broader role of dynamic pricing in retail food markets and concludes that retailers and consumers must work together to mitigate waste, and that retailers should adopt new ways to manage perishables. Chung also notes that dynamic pricing can create perverse incentives, such as customers strategically waiting for reductions, which is a constraint we keep in mind rather than a deal-breaker.

Subramanian et al. [4] describe food spoilage as a major global problem and propose a system combining IoT, molecularly-imprinted polymer sensors, and machine learning, reporting 95% accuracy and 93% precision when monitoring volatile organic compounds (VOCs). Their work shows that VOC-style sensing is viable for freshness, even though our hardware (a single MOX gas channel inside a BME688) is far less specific. Where their system aims at a single-device, high-confidence classification, we accept a coarser three-state output and rely on session-level baselining to compensate.

Albrecht et al. [1] propose a real-time dynamic shelf-life criterion for meat supply chains by combining sensory modelling and predictive microbiology. Although their domain differs from ours, their underlying claim, that freshness should be inferred from real-time, product-specific signals rather than from a fixed expiry date, is the same conceptual move we make for fruit, scaled down to a low-cost MOX-sensor setup. Taken together, the four references converge on three points that shaped our design: spoilage is a measurable signal at the IoT scale, dynamic pricing is the most plausible commercial vehicle for such a signal, and freshness is best treated as a real-time variable rather than as a printed expiry date.

3 Methodology

3.1 Field observation and cardboard prototype

We started with field observation. After noticing that supermarkets typically present fresh produce in modular cardboard boxes that staff swap quickly, we visited a local Spar store. Their fresh-produce shelving uses a clip rail along the front edge of each box, and any tag that obstructed that edge or required tools to remove would be incompatible with the existing workflow. This led to a clip-on smart tag concept that can be removed and reattached without disrupting box swaps. We then built a cardboard mock-up of a single shelf segment to fix dimensions and to test sensor and tag placement options before laser-cutting in plywood.

3.2 Iterative prototyping

Throughout the project we used iterative prototyping: build, test, analyse, refine. The choice was driven by two factors. First, the brief explicitly values process as well as the final artefact. Second,

real constraints made a single-shot design impractical. Components arrived late, the laser cutter was temporarily unavailable, and the wireless link between the ESP32 and the Raspberry Pi required several rounds of debugging before it became reliable. Each iteration cycle had the same shape: a build phase of one to three days, a quick informal test, and a short retrospective in which we recorded what worked and which open issues to tackle next.

The retail tag in particular went through two substantive design iterations, summarised in Figure 1. The first design was a single object that combined the e-ink display and the BME688 sensor into one piece, with the sensor extending out of the tag through a long cable that could be slipped into the produce box. We discarded this version because the cable made the device awkward to install and reposition every time staff changed boxes, and because routing a wire through a fresh-produce display was visually intrusive for customers. The second iteration, which is the final one, splits the system into two physically separate objects connected wirelessly. The sensing side is now an ESP32 that reads the BME688 and sends its readings to the server over Wi-Fi. The sensor itself is mounted at the tip of a 3D-printed arm that clips at the back of the produce box and rests on top of the produce, so that the only customer-facing surface is the e-ink shelf tag at the front. The arm ends in a small modelled fruit that holds the sensor, and the shape of that fruit is a small design detail that can be customised per product (a small orange for an orange box, a small apple for an apple box, and so on). The design detail keeps the device unobtrusive while giving each box a small visual cue tying the device to the produce it is monitoring.

3.3 Two-prototype strategy

As outlined in the introduction, we built two prototypes for different reasons, and not as two takes on the same product. The *retail prototype* is a research-style proof of concept that demonstrates the full pipeline; we make no claim that it is deployment-ready. The *educational prototype* is the version designed to actually be used, and doubles as a teaching artefact in schools. Sharing firmware between them made the trade-offs (network versus local, text versus colour, dashboard versus glance) explicit, and the parallel build gave us schedule resilience: when the retail build was blocked on networking, we could progress on the educational version, and the other way around.

3.4 Electronics selection

We selected components based on cost, power, and suitability: the **Adafruit 2.13" monochrome e-ink display** for its low power draw and suitability as a customer-facing label; the **Adafruit BME688 (STEMMA QT)**, which provides temperature, humidity, pressure, and a metal-oxide gas-resistance channel sensitive to reducing-gas VOCs in a single I²C package; the **Raspberry Pi Zero W** as the retail-side server and display node, hosting Flask and driving the e-ink panel directly; the **ESP32 Feather (Wi-Fi + Bluetooth)** as the wireless sensing node for the retail prototype; and the **ESP32-C3 DevKitM-1** with a **16-pixel NeoPixel RGBW ring** as the local-only sensing-and-display node for the educational prototype.

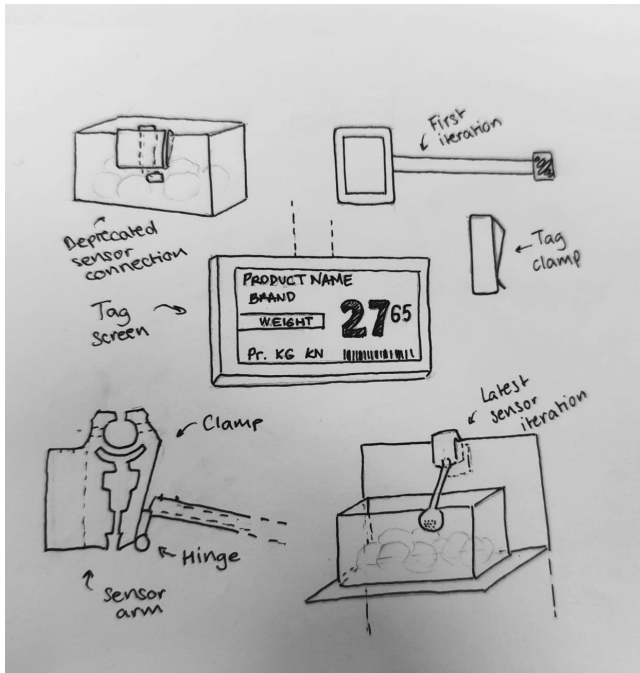


Figure 1: Sketch of the two main iterations of the retail tag. **Top:** the first design, a single object that combined the e-ink display and the sensor and routed a long cable into the produce box. **Bottom:** the final design, in which the sensing node is a separate wireless ESP32 and the sensor is mounted at the tip of a 3D-printed arm that clips at the back of the box.

4 Design and Fabrication

4.1 Retail prototype

The final retail tag is split into a sensing object and a display object. The sensing object is the 3D-printed arm described above, which clips to the back of the produce box and lowers the BME688 onto the surface of the fruit (Figure 2). The arm was printed in PLA and went through several minor revisions to tune the clip geometry and the depth at which the sensor sits inside the modelled fruit at the tip. The display object is the customer-facing e-ink tag attached to the front of the box, which shows the current freshness state and any active discount. We fabricated the supporting plywood shelf by laser cutting interlocking panels from 6 mm plywood, sized from the cardboard mock-up's dimensions and reinforced with 3D-printed corner mounts.

4.2 Educational prototype

The educational version was designed under different constraints: no network, no server, and a non-technical audience. The hardware is an ESP32-C3 DevKitM-1, the same BME688 sensor, and a 16-pixel NeoPixel RGBW ring inside a translucent 3D-printed enclosure that diffuses the LED output into a single coloured glow (Figure 3). The output is a three-state colour code: green for fresh, yellow for ripening, and red for very ripe. The translucent enclosure was



Figure 2: Final retail-tag sensing arm. The arm clips at the back of the produce box and ends in a small modelled fruit that holds the BME688 and rests on top of the produce. The shape of the modelled fruit at the tip is customised per product.

printed with reduced wall thickness specifically to let the NeoPixel ring's light diffuse evenly.

4.3 Fabrication techniques

Two techniques carried most of the physical work. **Laser cutting** was used for the plywood retail shelf, with interlocking teeth so the panels could be assembled by hand. **3D modelling and printing**, in PLA for the first two iterations of each part, was used for the sensor arm, the ESP32 enclosures, the Raspberry Pi/e-ink display housing, and the translucent educational enclosure. We tested clip-on geometry and tag-to-shelf fit by printing small fragments rather than full housings, which saved both time and material.

5 Implementation

5.1 System architecture

The retail prototype is a small networked pipeline. The BME688 talks to the ESP32 over I²C; the ESP32 posts JSON updates to the Raspberry Pi Zero W over Wi-Fi; the Pi runs Flask, applies the



Figure 3: Educational prototype. An ESP32-C3 with a BME688 and a 16-pixel NeoPixel RGBW ring sits inside a translucent 3D-printed enclosure. The whole enclosure glows green, yellow, or red according to the baseline-relative classification, with no on-screen text required.

classification logic, drives the e-ink shelf tag through its SPI interface, and serves the back-office dashboard over the local network. The educational prototype collapses the same pipeline into a single MCU. The ESP32-C3 reads the sensor over I²C, applies the same baseline-relative classifier in firmware, and drives the NeoPixel ring directly over GPIO using the WS2812 protocol.

5.2 Retail and educational firmware

The retail-side firmware on the ESP32 reads the BME688 over I²C and posts JSON updates to the Pi. The Pi runs a Flask application that exposes four routes (/store, /dashboard, /offers, /history) representing the customer-facing storefront, a back-office monitoring view, the active freshness-aware offers, and a record of past readings. A separate process drives the e-ink shelf tag from the latest classification, with refreshes throttled to respect the panel's update characteristics. The discount logic operates on baseline-relative quantities (§5.3) and classifies each box as *fresh*, *use-soon*, or *discount*; the corresponding state is then shown both on the e-ink tag (Figure 4) and on the dashboard. The educational firmware on

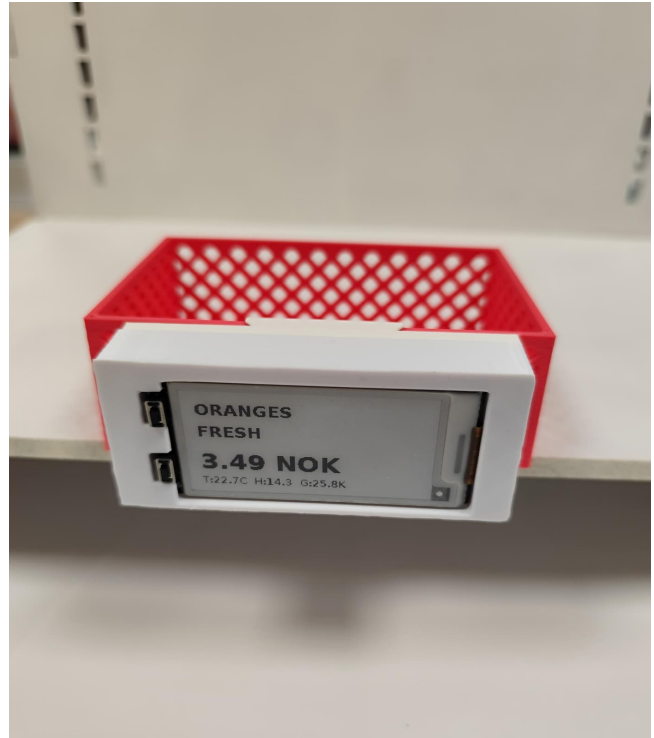


Figure 4: The customer-facing e-ink shelf tag, driven by the Raspberry Pi Zero W. The tag shows the current freshness state of the corresponding produce box and any active freshness-aware discount.

the ESP32-C3 reads the same sensor and uses the same baseline-relative classification, but maps the three states directly to colours on the NeoPixel ring (green, yellow, red). It has no Wi-Fi, no server, and no persistent state beyond the current session.

5.3 Baseline-relative firmware logic

Both prototypes share a firmware philosophy that we converged on after the orange experiment in §6. At session start, the firmware allows the BME688 to warm up in ambient air, then averages a short window of samples to define a per-session baseline for temperature, humidity, and gas resistance. Subsequent samples are interpreted relative to this baseline, as a humidity rise (in percentage points) and a gas-resistance drop (as a percentage of baseline). The classifier combines the two: a small humidity rise with no meaningful gas drop is *fresh*; a larger humidity rise with a moderate gas drop is *use-soon* (or *ripe* for the educational prototype); a large humidity rise combined with a large gas drop is *discount* (or *very ripe*). We deliberately avoid claiming an absolute freshness measurement. The metal-oxide channel responds to total reducing-gas load rather than to a specific molecule, the sensor drifts during warm-up, and ambient conditions vary between sessions, so a relative reading against a session baseline is the most defensible signal we can extract from this hardware.

5.4 Open-source release

The full project is released as an open-source repository, available at <https://github.com/natalialoprom/DIGIFAB>. The repository is organised by deployment context, with separate folders for the retail prototype (retail_prototype/), the educational prototype (educational_prototype/), the sensor experiment (experiments/), the laser-cut and 3D-printed source files (fabrication/), the paper itself (paper/), and supporting documentation (docs/). Each prototype folder contains the firmware sources, the CAD files, and a step-by-step README that walks the reader through wiring, flashing, calibration, and use. The intent of these instructions is that someone with no prior involvement in our project should be able to follow them from start to finish and end up with a working device, for either the retail or the educational version.

The educational prototype is the most direct beneficiary of the release. Schools and maker spaces interested in food-waste awareness or in basic embedded-electronics teaching can clone the repository and build the device with very little additional effort. Suggested extensions, such as repurposing the device for a different fruit by reprinting the modelled fruit at the tip of the sensor arm or by adjusting the colour thresholds, are documented as exercises so that the build can grow with the user. Beyond the build itself, the experiments/oranges/ folder contains the raw CSV from the experiment in §6, the cleaned dataset, and the analysis scripts that produced the figures in this paper, so the experimental claims here are reproducible from the repository alone.

We release the project openly because we expect that to be the most direct way for it to have practical impact. The retail concept on its own does not reduce food waste, since we cannot deploy it in any actual store. The educational version, on the other hand, can be picked up by another student team or by a teacher and used as it is, or extended to fit a particular school context. Treating the release as part of the deliverable, and not as an afterthought, is the way we aligned the project with SDG 12 in practice rather than only in framing.

6 Sensor Experiment

To calibrate the classification thresholds we ran a controlled three-condition experiment with the ESP32 Feather and BME688, comparing ambient air, freshly purchased oranges, and overripe oranges. The sensor was wired over I²C and an Arduino sketch streamed CSV samples over the serial monitor every ~2.4 s.

6.1 Procedure

The procedure was: (1) warm the sensor up in room air for ~23 minutes; (2) run a 20-sample baseline calibration in clean ambient air, yielding $T_b = 26.04$ °C, $H_b = 26.49$ %, and $G_b = 32.20$ k Ω ; (3) record a short ambient reference window (49.7 s, $n = 22$); (4) place fresh oranges near the sensor and record for 11.6 minutes ($n = 294$); (5) replace them with overripe oranges and record for 2.8 minutes ($n = 73$). We kept positioning consistent across the three phases and avoided breathing directly on the sensor. The enclosure was passively ventilated, on the assumption that at the slow natural rates of fruit change observed in the experiment, passive ventilation was sufficient to keep the local microenvironment representative.

Table 1: Mean sensor values per experimental segment. “Gas drop” is the resistance drop relative to the per-session baseline ($G_b = 32.20$ k Ω); “ Δ Hum” is the humidity rise in percentage points relative to $H_b = 26.49$ %.

Metric	Ambient	Fresh	Overripe
Temperature (°C)	26.01	24.27	24.64
Humidity (%)	26.57	30.84	34.89
Gas resistance (k Ω)	32.46	30.92	15.45
Gas drop (% of baseline)	-0.82	3.98	52.0
Δ Hum (pp)	+0.08	+4.34	+8.40
n / duration (s)	22 / 50	294 / 694	73 / 171

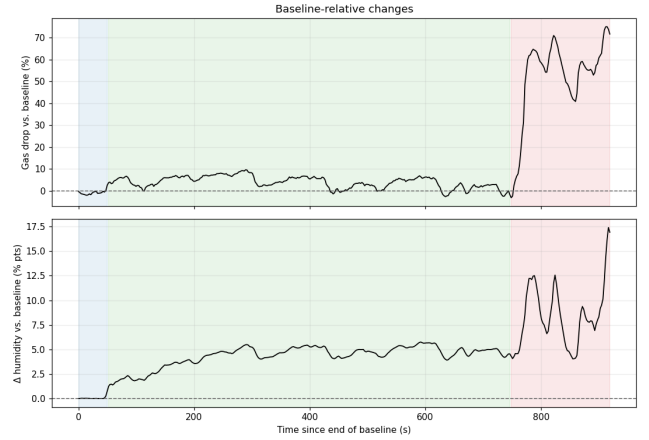


Figure 5: Baseline-relative signals across the three experimental phases. Top: gas-resistance drop as a percentage of baseline. Bottom: humidity rise in percentage points relative to baseline. The shaded regions correspond to the ambient reference, fresh-orange, and overripe-orange phases. Both signals separate the three phases monotonically, with a much larger relative excursion in the gas channel.

6.2 Results

Table 1 summarises the per-segment means, and Figure 5 shows the two baseline-relative signals (gas drop and humidity rise) used directly by the classifier. Both fruit conditions raise local humidity and lower gas resistance relative to ambient, but the magnitudes are roughly an order of magnitude apart. Fresh oranges produced a mean humidity rise of +4.34 percentage points (pp) and a mean gas drop of ~4 % relative to baseline. Overripe oranges produced a mean humidity rise of +8.40 pp (peak +17.4 pp) and a mean gas drop of ~52 % (peak ~75 %); the 50 % gas-drop threshold was crossed within ~26 s of the phase change. The overripe trace also showed strong oscillations between roughly 8 and 33 k Ω , consistent either with pulsed VOC release from the rotting fruit or with the BME688’s hot-plate duty-cycle producing recovery ramps when concentrations change rapidly.

6.3 Implications for the firmware

Two observations shaped the firmware in both prototypes. First, gas resistance is a much more sensitive discriminator than humidity.

Humidity rises about 2× between fresh and overripe, while baseline-relative gas drop rises by more than 10× over the same step, so we weight gas drop more heavily in the classifier. Second, even fresh fruit produces a non-trivial humidity rise within the first minute, so the classifier needs to distinguish “fruit is present” from “fruit is spoiling”. We use a two-axis threshold on humidity rise and gas drop rather than a single combined score. The cross-phase temperature drift (~1.7 °C between ambient and fruit segments) is a confound for the temperature-dependent MOX channel, so in deployment we re-calibrate the baseline whenever ambient temperature drifts beyond a few tenths of a degree.

7 Discussion and Limitations

The experiment used a single specimen of each fruit condition, in one enclosure, and in one session, so the numerical thresholds we derived should be read as informative rather than generalisable. The three phases also differ in duration (~50 s, ~694 s, ~171 s), so per-segment statistics are not balanced trials. The MOX gas element responds to total reducing-gas load rather than to a specific molecule, so we cannot attribute the gas drop to ethanol, ethylene, or any single VOC without auxiliary instrumentation. None of these caveats invalidates the qualitative ordering ambient < fresh < overripe, but they do mean that a deployed device should be presented as a comparative indicator rather than as an absolute spoilage meter.

The fabrication and integration side had its own limits. Component lead times pushed the wireless-tag iteration later than planned, laser-cutter availability was intermittent during the build window, and the wireless link between the ESP32 and the Pi required several rounds of debugging before it became reliable. These are routine constraints for short-cycle digital-fabrication projects, but they materially shaped the scope of the final prototype, and they are part of the reason we pursued two prototypes in parallel rather than one polished build.

8 Conclusion and Future Work

We presented Smart Tag, a two-prototype digital-fabrication project framed around UN SDG 12 that combines low-cost sensing (BME688), embedded networking (ESP32 / ESP32-C3), local computation (Raspberry Pi Zero W running Flask), e-ink and NeoPixel output, and laser-cut plus 3D-printed enclosures. Both prototypes share a baseline-relative firmware philosophy, calibrated against a controlled three-condition orange experiment in which baseline-relative gas drop separated fresh from overripe fruit by roughly an order of magnitude.

The retail prototype shows that the full pipeline, from wireless sensing to dashboard and freshness-aware discounting, can be built on student hardware. We are deliberate about not over-claiming, since actually deploying such a device in a supermarket is a procurement, certification, and liability problem more than a sensing problem. The educational prototype is the version we believe can have impact today. It is small, cheap, fully local, and well-documented enough that a school canteen can use it as a food-waste awareness object and a teacher can use it as a hands-on introduction to embedded electronics, sensor calibration, and digital fabrication. Releasing the full project as an open-source repository is the mechanism by

which we expect that impact to actually reach schools beyond our own course.

Future work falls into three areas. On the experimental side, we would like to repeat the orange experiment with multiple specimens per condition and a tighter enclosure so that the thresholds can be reported with confidence intervals, and to extend the protocol to other fruit and to vegetables typical of school canteens. On the firmware side, we plan to replace the hand-tuned thresholds with a small machine-learning model trained on the same baseline-relative features, and to investigate a rolling-baseline scheme that does not require explicit calibration at session start. On the educational side, we want to package the build as a structured workshop, with a sequence of short exercises that walks a class from soldering the sensor to interpreting their own captured data, so that the open-source release becomes a complete teaching unit rather than only a code drop.

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References

- [1] Antonia Albrecht, Marlene Mittler, Markus Hebel, Christian Waldhans, Ulrike Herbert, and Judith Kreyenschmidt. 2021. A Multi-Model Approach to Implement a Dynamic Shelf Life Criterion in Meat Supply Chains. *Foods* 10, 11 (2021), 2740. <https://doi.org/10.3390/foods10112740>
- [2] J. Chung. 2019. Effective Pricing of Perishables for a More Sustainable Retail Food Market. *Sustainability* 11, 17 (2019), 4762. <https://doi.org/10.3390/su11174762>
- [3] K. Sathyabama, A. P. A. Raj, and R. Gukan. 2024. AI-Driven Expiry Prediction and Dynamic Pricing System for Retail Inventory Management Utilizing ARIMA and LSTM Models. In *2024 International Conference on Emerging Research in Computational Science (ICERCS)*. <https://doi.org/10.1109/ICERCS63125.2024.10895116>
- [4] R. R. Subramanian, G. Charan, G. V. Krishna, G. Sashavali, and G. S. Yeswanth. 2025. Smart Packaging: IoT-Integrated Sensors for Food Freshness. In *2025 International Conference on Computational Robotics, Testing and Engineering Evaluation (ICCRTEE)*. <https://doi.org/10.1109/ICCRTEE64519.2025.11052946>

Use of AI Tools

During the preparation of this paper we used generative AI tools to assist in drafting and refining parts of the text. All AI-assisted content was thoroughly reviewed against our own experimental data, design notes, and source code, and was edited where necessary to remove inaccuracies or inconsistencies. The authors take full responsibility for the final content of the paper.